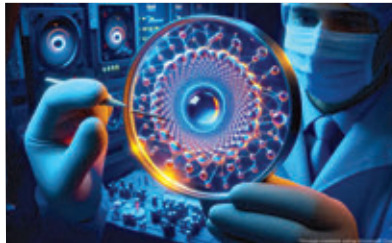


■ OPTICAL TECHNOLOGY

Scientists develop thinnest lens ever

Scientists have developed the thinnest lens ever made, measuring just three atoms thick, thanks to quantum phenomena. This innovative lens allows most wavelengths of light to pass through, making it highly promising for applications in optical fibre communication and devices like augmented reality glasses. The researchers who developed the lens, from the University of Amsterdam in the Netherlands and Stanford University in the US, believe their innovation will advance research into lenses of this type and miniature electronic systems, reports ScienceAlert. "The lens can be used in



applications where the view through the lens should not be disturbed, but a small part of the light can be tapped to collect information," said Jorik van de Groep, a nanoscientist at the University of Amsterdam.

Instead of bending light through refraction with a transparent material's

curved surface, incoming waves are focused by a series of grooved edges using diffraction.

This technology, known as a Fresnel lens or zone plate lens, has been utilised for centuries to create thin, lightweight lenses, such as those found in lighthouses. To enhance the technique with a quantum boost, the research team etched concentric rings into a thin layer of tungsten disulfide (WS₂), a semiconductor. When WS₂ absorbs light, its electrons move in a precise way, creating gaps that can be considered particles in their own right.

<https://bit.ly/45D7C1T>

■ MICROPROCESSORS

Researchers develop Superman-inspired 'Imager Chip'

Imagine a world where your smartphone could see through walls, detect hidden objects inside packages, or reveal concealed structures with just a click. This futuristic vision, reminiscent of Superman's X-ray vision, seems to be inching closer to reality. Researchers have developed an imager chip that, when integrated into mobile devices, could revolutionise the way we perceive and interact with our surroundings. Researchers from The University of Texas at Dallas and Seoul National University have developed an imager chip inspired by Superman's X-ray vision. This innovation aims to enable mobile devices to detect objects concealed within packages or behind walls.

Smartphones equipped with this chip could potentially locate studs, wooden beams, wiring behind walls, cracks in pipes, and

even identify contents within envelopes and packages. This technology also could have medical applications.

The researchers initially showcased the imaging technology in a study from 2022. Their most recent paper, published in the March issue of IEEE Transactions on Terahertz Science and Technology, reveals how they tackled a major hurdle: shrinking the technology to fit handheld mobile devices while enhancing image quality. "This technology is like Superman's X-ray vision. Of course, we use signals at 200 gigahertz to 400 gigahertz instead of X-rays, which can be



harmful," said Dr. Kenneth K. O, director of the Texas Analog Center of Excellence (TxACE) and the Texas Instruments Distinguished University Chair in the Erik Jonsson School of Engineering and Computer Science.

<https://bit.ly/3VxMrTi>